

SAI VENKATA NAGENDRA ABHAY RAJ GURRAM

Jersey City, NJ, United States

+1 815-618-1777 | abhayrajgurram13@gmail.com

linkedin.com/in/abhaygurram | github.com/Abhayraj1307

PROFESSIONAL SUMMARY

Recent Master's graduate in Business Analytics and Artificial Intelligence with hands-on experience in machine learning, NLP, search-oriented workflows, embeddings, clustering, model evaluation, data pipelines, and AI-driven analytical systems. Skilled in Python, NumPy, Pandas, Scikit-learn, Git, Streamlit, FastAPI, and SQL, with exposure to PyTorch, TensorFlow, Transformers, LLMs, RAG pipelines, and vector search concepts. Strong interest in building production-oriented ML systems for search, ranking, recommendations, document extraction, and domain-specific information retrieval.

TECHNICAL SKILLS

Programming & Data: Python, SQL, NumPy, Pandas, Data Cleaning, Data Preprocessing, Feature Engineering, Data Transformation

Machine Learning: Scikit-learn, Supervised Learning, Unsupervised Learning, Classification, Clustering, Recommendation Systems, Model Training, Model Evaluation, Cross-Validation, Regularization, Bias-Variance Tradeoff

Deep Learning & NLP: PyTorch Exposure, TensorFlow Exposure, Transformers, BERT, Tokenization, Embeddings, NLP Pipelines, Text Classification, Information Extraction, Document Understanding

Search & AI Systems: Dense Vector Embeddings, Hybrid Search Concepts, Retrieval and Ranking, Re-ranking Concepts, RAG Pipelines, Vector Databases Exposure, FAISS Exposure, Prompting, LLM Workflows

Engineering & Deployment: Git, Version Control, Modular Code, Reproducible Workflows, Streamlit, FastAPI, API Integration, Testing Awareness, Debugging, Documentation

Visualization & BI: Matplotlib, Plotly, Power BI, Tableau, Analytical Dashboards, Reporting Metrics

PROJECT EXPERIENCE

NovaMind – AI Decision Intelligence Platform

- Built an AI-enabled decision support platform integrating NLP workflows, retrieval-style reasoning, structured analytical pipelines, and real-time outputs.
- Designed modular Python workflows for data preprocessing, model-assisted analysis, evaluation, and decision generation.
- Developed Streamlit-based interfaces to visualize analytical outputs, model summaries, and business decision-support results.
- Applied RAG-style concepts, prompt workflows, and structured output generation for domain-specific information retrieval and analysis.

Twitter Sentiment Analysis using BERT

- Developed a Transformer-based NLP classification workflow using BERT for sentiment analysis on social media text data.
- Performed text preprocessing, tokenization, embedding generation, model evaluation, and classification performance reporting.
- Built visualization outputs to communicate sentiment trends, model behavior, and analytical insights.
- Applied Python-based ML experimentation using reproducible notebooks and structured evaluation workflows.

Formula 1 Strategy Analysis using Clustering

- Applied unsupervised machine learning to identify race strategy patterns and performance clusters from structured datasets.
- Conducted preprocessing, feature engineering, clustering analysis, and model evaluation using Python, Pandas, NumPy, and Scikit-learn.
- Built analytical dashboards and visualizations to compare performance segments, trends, and strategy outcomes.

- Translated model outputs into actionable insights supporting data-driven performance analysis.

Rationale Agent – AI Incident Intelligence System

- Developed an AI-assisted analysis system for incident investigation, root-cause reasoning, evidence extraction, and structured reporting.
- Built modular pipelines for input parsing, analytical reasoning, hypothesis comparison, and output validation.
- Applied model-assisted workflows to support debugging, evaluation, and decision-ready reporting for operational problems.

Netflix Recommendation System Optimization

- Conducted exploratory data analysis, clustering, forecasting, and engagement pattern analysis on structured entertainment data.
- Built Python-based workflows for preprocessing, feature analysis, visualization, and recommendation insight generation.
- Created dashboard-style reporting outputs to summarize user engagement patterns, content trends, and recommendation opportunities.

Historacle AI – Historical Intelligence Dashboard

- Built an interactive dashboard for structured data exploration, trend analysis, and insight generation.
- Developed workflows for organizing, visualizing, and interpreting domain-specific information through analytical interfaces.

PROFESSIONAL EXPERIENCE

Quality Assurance Intern

Jan 2022 – Jun 2022

Laurus Labs, Visakhapatnam, India

- Performed structured data validation, documentation review, and process verification within regulated operational workflows.
- Analyzed records and identified inconsistencies to support corrective actions, reporting accuracy, and process reliability.
- Collaborated with cross-functional teams to maintain documentation quality, operational standards, and data integrity.

EDUCATION

Stevens Institute of Technology

May 2026

Master's in Business Analytics & Artificial Intelligence | GPA: 3.51

Relevant Coursework: Machine Learning, Data Mining, Web Mining, Data Management, Optimization & Process Analytics, Multivariate Data Analysis

GITAM Deemed University

Bachelor's in Pharmacy

CERTIFICATIONS & ADDITIONAL INFORMATION

Apple Search Ads Certification | Bank of America Investment Banking Simulation

Publication: UV Analysis of Ritonavir | Languages: English, Telugu